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Taste of Thoughts, Homes of Ideas

Abstract

Language translation can be considered as a type of transformation. In which an idea is given and presented in a new linguistic form. Added to this is the notion of metaphor. With this figure of speech, language users can talk of a thing by referring to another thing which shares some features in common. These linguistic activities demonstrate how languages by nature can metamorphose. Building upon these acts, this NLP project attempts to replicate this nature of human language by transforming a textual representation of general thought into a textual form of either sensory description of food or description of spatial property. I implemented this project in Python, using libraries such as Spacy, Tracery, and Markovify. The program operates by these three steps: extracting, mapping, and generating. NLP tasks involved in this procedure are linguistic analysis, vector embedding, and text generation.

Introduction

This project stems from my interest in Walter Benjamin's ideas on translation. As we may commonly conceive of translation as an act to communicate a message from a source language into a target language, the action is arguably not a mere repetition but rather a kind of transformation and creation that amplifies "the echo of the original" (Benjamin, 1996, p. 258). Taking a closer look, translation comprises multiple processes of changing an original text and giving it a new literary form, and these new lexical or stylistic forms are attached by values which will vary by time or culture, as Benjamin says:

> For in its afterlife — which could not be called that if it were not a transformation and a renewal of something living — the original undergoes a change. Even words with fixed meaning can undergo a maturing process. The obvious tendentiousness of a writer's literary style may in time wither away, only to give rise to immanent tendencies in the literary creation. (p. 256)

Moreover, it can be said that translation is also production. A translation enables an intended thought of a person to be conveyed in a new form, which in turn highlights the original form the idea inhabits, as quoted: "[a] real translation is transparent; it does not cover the original, does not block its light, but allows the pure language, as though reinforced by its own medium, to shine upon the original all the more fully" (p. 260). With these new perspectives on translation, I found the inspiration for this project.

Another idea this project is based on involves metaphor. The term can be defined as "a figure of speech in which a word or phrase literally denoting one kind of object is used in place of another to suggest a likeness between them" (Martinich, 1998). It fascinates me how we can talk of something in a non-literal way by referring to something else that may belong to a different category of concept to the former. To illustrate, in English, abstract notions like thoughts or ideas are often described as if they are food, e.g. "sweet idea" or "spicy opinion". Additionally, given the definition, a parallel between translation and metaphor can be drawn. As the creative use of language is to put a meaning in a form of another thing that shares some similar features, creation and expression of metaphors is arguably a kind of translation.

Considering these ideas altogether, they can be formulated into a question, as follows: how can forms of thought be transformed or translated into forms of sensation that are relevant to everyday life? To limit the scope of this project, one of these sensory forms is description of food tasted. In addition to the metaphors of thoughts as food above, taste is also a site of knowledge which enables us to learn about things (Richardson, 2021). Although its epistemological power is often neglected, taste is a sense which connects ourselves to the world. Another form that is the target of this metamorphosis involves describing spaces or interior designs of home. There is a relation between home space and ourselves. Our internal thoughts are projected onto the space of home, as quoted here: "[a] home is not just the space in which things become subjects. It is also the place in which all things become extensions of our subjectivity." (Coccia, 2024, p. 71). These are two domains of language this project will work on, as they are activities close to our everyday life.

Given this question, this project is an attempt to actualise a NLP program that can translate or transform a textual representation of general thought into a textual form of either sensory description of food or description of spatial property. In the next section, I will give details on the workflow and tools used for this project. Next, I will describe the features of the program. Following this, I will narrate how I developed this application. Then, I will provide evaluations of the program's performance. Lastly, I will identify limitations of the work and offer suggestions for prospective projects.

Methodology

The program was developed in Python using PyCharm as the integrated development environment (IDE). To do the NLP tasks of this project, three relevant libraries were used. The library Spacy was employed to perform tokenisation and linguistic analysis on an input as well as to calculate similarity between tokens. For generating texts for an output, this part was done by usage of Tracery and Markovify libraries. The former generates texts with a set of given rules whereas the latter generates texts by shuffling sequences of linguistic units. Moreover, Emotion English DistilRoBERTa-base (Hartmann, 2022), a Transformer model for classifying emotions in English texts, was used to identify moods of an input text.

The principal workflow of this programme consists of three processes: **extracting**, **mapping**, and then **generating**. After receiving a textual input from the user, the program processes and extracts content words (i.e. nouns, verbs, and adjectives) that are the main themes of that input. By using the classifying model, the program also derives emotions or moods from that text. With this obtained information, the program maps the extracted themes to semantically similar words of a target category (i.e. food or furniture) by comparing each theme word to each word in that category and calculating a similarity score via word vectors. Likewise, the application also maps the derived emotions to words of a sensory category (i.e. flavor or spatial atmosphere). Given these mapped words and a set of rules designed beforehand, a description text is created. This text is then used for training a generative model and then combined with another model trained by a language corpus. Utilising this combined model, the program produces an output text, which its content parallels the input.

NLP techniques involved in these processes are linguistic analysis, vector embedding, and text generation. In linguistic analysis, there are two sub-tasks, tokenisation and part of speech tagging. This technique is employed to extract content words from an input text by selecting words of specific classes (i.e. nouns, verbs and adjectives). To do word mapping, vector embedding is utilised to compute semantic similarities between words for finding neighbor words or solving analogies. For text generation, this technique has subsequent tasks of creating a base text from Tracery's generative grammar and of making language models from this base text and a corpus text using Makovify. The models are then combined to generate an output text, containing theme words of the base and stylistics of the corpus.

Given time and resource constraints, the vocabularies for mapping source words to target words and the language corpora for training the language models in this project were all generated by ChatGPT. The words and sentences are stored in .txt files with white spaces separating one word or sentence from another.

Development Process

After coming up with the inspiration for the project, I began the development process by creating a prototype program. The code was at first implemented on Jupyter Notebook, so I could check how the code works line by line. Using existing code from the class and getting assisted by ChatGPT, I built functions that execute the three main tasks of this program. By testing the app's performance, the prototype can receive an input and successfully transform it into a sensory narration of food. This step is hence to test whether or not my idea can really work out and whether the algorithm designed initially can function with no crucial problems.

The prototype was commented by my instructor that the processes of translating thoughts into food and of translating thoughts into home objects should be differentiated — for the relationship between objects and abstract thoughts is presumably less straightforward — and that the output should vary more. With this feedback, I made four further adjustments. First, to handle the issue of translating thoughts into home objects, I added a semantic bridge of abstract concepts (e.g. hope, grief, etc.) in the process. This bridge makes the connection between the input and objects smoother by generalising the former to be closer to the latter. When content words are extracted from an input, the words are first mapped to the concepts. These mapped concepts are then mapped to the furniture words. Second, I augmented the size of a base text by including various sentence structures in Tracery's grammar, so the output can be various. Third, I had the program detect whether an input is heavily loaded by either food or object words. If one of these happens to be the case, the input will be translated into the language of its counterpart's domain. Lastly, to have more various outputs, I had the app select a corpus for blending with the base text, given the most dominant emotion of the input.

Program Design

The web app's interface is plain and simple. At the top of the page shows the title and welcoming message, briefly describing what this program does. Below are the instructions on how to use the app, telling the user to write down a text and to choose the domain of target language. Then follows the section where the user can now interact with the app. In this part, there is a text area widget where the user can input their text and a selection box where they can select a language domain. Under the text area is a text displaying word count of the input. At the bottom of the page is the button embedded with the function for generating an output text. Once the text is generated and displayed, a button which enables the user to copy the output appears, as well as a button embedded with the link to the survey. If there is no input text given, the app shows a message warning the user to enter some input into the textbox.

Evaluation

Now that the process of developing the program has been finalized, in this step, I aim to test and evaluate the program's performance, given different types of input. There are [...] testing cases in this report. The picture examples can be found in the later pages.

Case 1 - Input of less-than-20-words length

input:	I love you	
themes:	'love'	
emotions:	'joy'	
	culinary	interior
mapped1:	'strawberry'	'stand'
mapped2:	'sweet'	'joyful'
output	As it rests, the strawberry sharpens its sweet into a precise sequence. The strawberry balances sweet intensity with a sudden turn toward sweet.	The scale of the stand reveals a crisp joyful clarity.

In this case, I tested how the program works when the size of an input is small, containing less than 20 words. Given this type of input, the app can generate some outputs. The output of each domain also contains the mapped word. However, the app sometimes did not return any output text. This may be due to the small size of the input. I then adjusted the code to warn the user to click and generate an output again if the previous output is empty.

With small-size inputs, the program can still function, but it sometimes does not generate any output. The issue is to be resolved by generating an output text again.

Case 2 - Input of long length

input:	Lately I've been drifting through my days with a quiet restlessness, torn between the comfort of routine and the sharp pull of something unknown. I crave the safety of familiar streets and steady mornings, yet at night I imagine disappearing into a distant city where no one knows my name. There's a strange mix of gratitude and dissatisfaction inside me — a warmth for what I have, and a hunger for what I haven't dared to chase. Sometimes it feels like I am rooted and weightless at the same time, grounded by responsibility but haunted by possibility.
themes:	'drift', 'day', 'quiet', 'restlessness', 'tear', 'comfort', 'routine', 'sharp', 'pull', 'unknown', 'crave', 'safety', 'familiar', 'street', 'steady', 'morning', 'night', 'imagine', 'disappear', 'distant', 'city', 'know', 'strange', 'mix', 'gratitude', 'dissatisfaction', 'warmth', 'hunger', 'dare', 'chase', 'feel', 'root', 'weightless', 'time', 'ground', 'responsibility', 'haunt', 'possibility'
emotions:	'fear'

	culinary	interior
mapped1:	'marjoram', 'sage', 'bay', 'lavender', 'coffee', 'date', 'milk', 'orange', 'goose', 'duck', 'bean', 'tea', 'water', 'banana', 'butter'	'throne', 'heater', 'step', 'shade', 'alarm', 'safe', 'incense', 'column', 'vanity', 'stand', 'cushion', 'case', 'dome', 'beam', 'perch'
mapped2:	'intense'	'darkness'
output	Each swallow of custard grows more watery, echoing like a pulse racing in the throat. The milk organizes intense into a sustained calm. The banana holds a tactile memory of intense returns through the tea. The banana absorbs the intense touch lingers on the date. Steam from the thin bread curls softly like something unseen drawing nearer. Between texture and memory, the water is done, it becomes deeply intense. Long after the tea presents an exterior, it gradually reveals a layered intense architecture within the banana. The scent of sharp tea drifts upward like footsteps echoing behind you. The bay balances intense intensity with a intense edge that softens over time. The banana holds a tactile memory of intense surrounds the goose, and the intense slips away from the eerie onion lingers like breath caught before a scream.	The beam ages gracefully, enriching the darkness atmosphere unfolds in controlled stages. In the staircase, every table seems shadowed, like a whisper moving through empty halls. The heater reflects a quiet discourse on darkness within its geometry. In the staircase, the sofa turns the bathroom listens like breath held before a sound. Material textures within the dome retains its structural clarity, the darkness atmosphere. The tactile quality of the dome echo a subtle darkness rhythm.

In this case, the input's size was larger, containing 97 words. The app successfully functioned and generated output texts in both domains. The outputs contained the mapped words, with the moods ('intense' and 'darkness') parallel the source text. Still, the lengths of the texts are ostensibly different. This might be caused by the length of template sentences in Tracery grammar. As it did not demonstrate any crucial issue, I did not make any change.

When the input got larger, the program still successfully functioned and produced more fruitful output texts. The difference in the output texts may be a bit distracting, but I did not consider this issue to be a big problem.

Case 3 - Input heavily loaded with food words

input: Yesterday I prepared a large dinner with rice, pasta, bread, butter, cheese, milk, cream, yogurt, and eggs. I roasted chicken with garlic, onion, pepper, salt, basil, rosemary, and thyme, and served it with tomato soup, grilled salmon, fried tofu, and steamed vegetables like broccoli, carrots, potatoes, spinach, and mushrooms. For dessert, we had chocolate cake, vanilla ice cream, strawberry jam, apple pie, banana pudding, and honey with almonds and walnuts. I also made a fresh salad with lettuce, cucumber, avocado, olive oil, lemon juice, and vinegar, alongside a bowl of noodles with beef, shrimp, chili sauce, soy sauce, and sesame seeds. To drink, we had coffee, tea, orange juice, and a smoothie made from mango, pineapple, and coconut milk. The table was completely covered with plates of sushi, curry, dumplings, sausage, bacon, and sandwiches filled with ham and cheese.

themes:	'yesterday', 'prepare', 'large', 'dinner', 'rice', 'pasta', 'bread', 'butter', 'cheese', 'milk', 'cream', 'yogurt', 'egg', 'roast', 'chicken', 'garlic', 'onion', 'pepper', 'salt', 'basil', 'rosemary', 'thyme', 'serve', 'tomato', 'soup', 'grill', 'salmon', 'fry', 'tofu', 'steam', 'vegetable', 'broccoli', 'carrot', 'potato', 'spinach', 'mushroom', 'dessert', 'chocolate', 'cake', 'vanilla', 'ice', 'cream', 'strawberry', 'jam', 'apple', 'pie', 'banana', 'pudding', 'honey', 'almond', 'walnut', 'fresh', 'salad', 'lettuce', 'cucumber', 'avocado', 'olive', 'oil', 'lemon', 'juice', 'vinegar', 'bowl', 'noodle', 'beef', 'shrimp', 'chili', 'sauce', 'soy', 'sauce', 'sesame', 'seed', 'drink', 'coffee', 'tea', 'smoothie', 'mango', 'pineapple', 'coconut', 'milk', 'table', 'cover', 'plate', 'sushi', 'curry', 'dumpling', 'sausage', 'bacon', 'sandwich', 'fill', 'cheese'
emotions:	'neutral', 'disgust'

	culinary	interior
mapped1:	-	'dome', 'painting', 'heater', 'shade', 'throne', 'step', 'arrangement', 'incense', 'portfolio', 'globe', 'stand', 'cushion', 'case', 'sheet', 'range', 'chrome', 'blender'
mapped2:	-	'balanced', 'grand', 'echoing'
output	-	Over time, the dome reveals a crisp grand clarity, while evening softens it into reflection. As one passes the heater, the room into a quiet meditation. The range filters sensory input into a tactile experience of dwelling. Over time, the globe projects a subtle grand rhythm. The shade absorbs the room's balanced mood. Through texture and form, the case in crisp grand clarity, while evening softens it into reflection. Each approach to the cushion presents a restrained exterior, it gradually reveals a balanced clarity. Seasonal light transforms the heater reframes the step scatter a faint grand shimmer across the blender, intensifying its balanced precision. Morning reveals the case negotiates a balanced calm. The cushion balances echoing presence accumulates with time. The blender feels like a slow philosophical argument. With repeated use, the arrangement establishes a rhythm of grand meaning. Against the wall, the globe echo a subtle echoing rhythm. Material textures within the domestic landscape. The arrangement feels like a slow philosophical argument.

For this case, I tested how the app operates when the input is heavily loaded with words about food. The input text contains a lot of words about food. After commanding the program to generate an output text, although the culinary mode was selected, the program displayed a text saying: "Food-heavy input detected. Translating into interior space". The text generated contained words mapped into the vocabulary of home objects.

For this type of input, the program also successfully operated and generated outputs.

Case 4 - Input heavily loaded with home object words

input:	This morning I walked through my apartment and noticed how every room was filled with furniture and small home objects. In the living room there was a sofa, armchair, coffee table, bookshelf, cabinet, television, floor lamp, and rug placed beside a wooden bench and a glass side table. The dining area had a large dining table surrounded by chairs, a cupboard filled with plates, bowls, cups, glasses, forks, knives, spoons, and a ceramic vase. In the kitchen I saw a refrigerator, oven, stove, microwave, kettle, toaster, cutting board, frying pan, saucepan, blender, and dish rack sitting next to the sink and faucet. On the counter there were jars, containers, a fruit bowl, and a basket. The bedroom contained a bed, mattress, pillow, blanket, wardrobe, dresser, mirror, bedside table, alarm clock, reading lamp, and a wooden chair near the window. In the bathroom there were shelves with towels, soap, shampoo bottles, toothbrushes, toothpaste, a hair dryer, and a laundry basket. Everywhere I looked there were objects like lamps, cushions, curtains, carpets, stools, cabinets, drawers, picture frames, plants, and decorative ornaments scattered throughout the house.	
themes:	'morning', 'walk', 'apartment', 'notice', 'room', 'fill', 'furniture', 'small', 'home', 'object', 'living', 'room', 'sofa', 'armchair', 'coffee', 'table', 'bookshelf', 'cabinet', 'television', 'floor', 'lamp', 'rug', 'place', 'wooden', 'bench', 'glass', 'table', 'dining', 'area', 'large', 'dining', 'table', 'surround', 'chair', 'cupboard', 'fill', 'plate', 'bowl', 'cup', 'glass', 'fork', 'knife', 'spoon', 'ceramic', 'vase', 'kitchen', 'see', 'refrigerator', 'oven', 'stove', 'microwave', 'kettle', 'toaster', 'cut', 'board', 'fry', 'pan', 'saucepan', 'blender', 'dish', 'rack', 'sit', 'sink', 'faucet', 'counter', 'jar', 'container', 'fruit', 'bowl', 'basket', 'bedroom', 'contain', 'bed', 'mattress', 'pillow', 'blanket', 'wardrobe', 'dresser', 'mirror', 'bedside', 'table', 'alarm', 'clock', 'read', 'lamp', 'wooden', 'chair', 'window', 'bathroom', 'shelf', 'towel', 'soap', 'shampoo', 'bottle', 'toothbrush', 'toothpaste', 'hair', 'dryer', 'laundry', 'basket', 'look', 'object', 'lamp', 'cushion', 'curtain', 'carpet', 'stool', 'cabinet', 'drawer', 'picture', 'frame', 'plant', 'decorative', 'ornament', 'scatter', 'house'	
emotions:	'disgust', 'neutral', 'surprise'	
	culinary	interior
mapped1:	'espresso', 'date', 'parsley', 'cream', 'cappuccino', 'water', 'smoothie', 'grape', 'pork', 'wine', 'butter', 'coffee', 'pumpkin', 'goose', 'bean', 'tea', 'soda', 'lavender', 'milk', 'duck', 'cherry', 'blackberry', 'bay', 'orange', 'chicken'	-
mapped2:	'tender', 'bitter', 'sweet', 'solid'	-
output	The curdled edge of the duck carries a sustained solid clarity. Steam from the putrid clam curls upward like breath trapped in a sealed room. The aftertaste of butter turns overripe, blooming like flesh surrendering to time. A smear of musty garlic lingers like decay hiding beneath sweetness, unwelcome and persistent. The putrid edge of the garlic breaks apart like sweetness turned traitor, unwelcome and persistent. Each bite of oyster tastes fungal, unfolding like sweetness turned traitor. Each bite of fish tastes spoiled, unfolding like sweetness turned traitor, pressing against the senses. The aftertaste of butter turns cloying, blooming like something left too long in the dark. The mouthfeel of the cream into a precise sequence. Long after the soda into a single moment. Between texture	-

	and memory, the water absorbs heat, intensifying its solid gravity. Between texture and memory, the soda into a more introspective tender tone. Between texture and memory, the soda into a precise sequence. Steam from the musty cheese curls upward like a bloom collapsing into mold. A layer of tender remains. The overripe edge of the yogurt feels curdled, sliding across the palate like decay hiding beneath sweetness.

Similar to the third case, I aimed to test how the program functioned when an input was heavily loaded with words about home objects. The test input contained a lot of such words. After commanding the program to generate an output text, although the interior mode was selected, the program displayed a text saying: “Space-heavy input detected. Translating into culinary taste”. The text was successfully generated and contained mapped words.

For this type of input, the program also successfully functioned and produced outputs.

Case 5 - Input of less-than-20-words length, deployed on Streamlit

input:	I love you	
themes:	'love'	
emotions:	'joy'	
	culinary	interior
mapped1:	'passionfruit'	'comforter'
mapped2:	'gentle'	'joyful'
output	The passionfruit absorbs heat, intensifying its gentle into a composed gentle calm. The aftertaste of the passionfruit, thick with gentle resonance.	As evening falls, the comforter in crisp joyful clarity. Under morning light, the comforter accumulates a deeper layer of joyful throughout the interior.

After having tested how the app operated on my local computer, I then proceeded to deploy the program on Streamlit. A few small changes were made to make the app executable on the platform, e.g. adding a requirement file and changing how the emotion classifier returns a result to fit the new version of transformers. The app can be now used on Streamlit.

To test how this web app works, I repeated the testing cases I did for my local app. In this case, I entered the same small input (i.e. “I love you”). The app successfully produced texts containing the mapped words. Yet, the results were inconsistent with the results of the first case. The content word ‘love’ was translated this time as ‘passionfruit’, not ‘strawberry’ as in the first case. Still, these new words did not change the sentiments of the texts.

In this web app version, the translation tool still functioned and successfully generated outputs. However, the new words that the original content words were mapped to may be different from the food or object words mapped in the local app version. This technical issue is beyond my current knowledge, and the code for the app and the function used in both versions are the same. I assumed that this may be accounted for by how the Python interpreter operates differently in Streamlit. Nevertheless, by my judgment, this is not a crucial issue and will be left unsolved as the output texts still conveyed the same moods.

Case 6 - Input of long length, deployed on Streamlit

input:	Lately I've been drifting through my days with a quiet restlessness, torn between the comfort of routine and the sharp pull of something unknown. I crave the safety of familiar streets and steady mornings, yet at night I imagine disappearing into a distant city where no one knows my name. There's a strange mix of gratitude and dissatisfaction inside me — a warmth for what I have, and a hunger for what I haven't dared to chase. Sometimes it feels like I am rooted and weightless at the same time, grounded by responsibility but haunted by possibility.	
themes:	'drift', 'day', 'quiet', 'restlessness', 'tear', 'comfort', 'routine', 'sharp', 'pull', 'unknown', 'crave', 'safety', 'familiar', 'street', 'steady', 'morning', 'night', 'imagine', 'disappear', 'distant', 'city', 'know', 'strange', 'mix', 'gratitude', 'dissatisfaction', 'warmth', 'hunger', 'dare', 'chase', 'feel', 'root', 'weightless', 'time', 'ground', 'responsibility', 'haunt', 'possibility'	
emotions:	'fear'	
	culinary	interior
mapped1:	'water', 'milkshake', 'persimmon', 'dragonfruit', 'coffee', 'passionfruit', 'bulgur', 'bay', 'smoothie', 'turnip', 'starfruit', 'tempeh', 'vanilla', 'date', 'clementine', 'orange'	'downlight', 'case', 'threshold', 'stand', 'spotlight', 'partition', 'extension', 'comforter', 'vanity', 'valance', 'centerpiece', 'beam'
mapped2:	'stronger', 'tenderest'	'classic', 'darkness', 'texture'
output	Because the vanilla absorbs the tenderest unfolds in controlled stages. At first restrained, the coffee sharpens its stronger identity. The watery edge of the apple turns watery, hovering like eyes watching from the eerie onion curls softly like footsteps echoing behind you, leaving the mouth uncertain. The coffee, strangely sour, spreads across the palate like eyes watching from the body of the fish grazes the senses like a pulse racing in the throat. Steam from the date. The clementine redefines what tenderest can be. Between texture and memory, the smoothie speaks in tenderest tones.	The rug unsettles the bathroom, watchful and tense like a whisper moving through phases of darkness. The trembling surface of the spotlight into a sustained calm. While the centerpiece reframes the interior within texture perspective. The valance, initially marked by darkness, as the space a lasting texture depth. The faded surface of the partition accumulates a deeper layer of darkness experience. The centerpiece filters sensory input into a marker of darkness restraint. The partition sharpens visual focus within a timeless classic field. The strained surface of the spotlight anchors the room in a distinctly darkness atmosphere.

For this test case, I entered the longer text into the web app. While the mapped words were different from the second case, they share some same words: 'water', 'coffee', 'bay',

‘dae’ in the food domain; ‘case’, ‘beam’, ‘vanity’ in the design domain. Overall, the program still functioned and successfully generated the outputs containing those mapped words.

Case 7 - Input heavily loaded with food words, deployed on Streamlit

input:	Yesterday I prepared a large dinner with rice, pasta, bread, butter, cheese, milk, cream, yogurt, and eggs. I roasted chicken with garlic, onion, pepper, salt, basil, rosemary, and thyme, and served it with tomato soup, grilled salmon, fried tofu, and steamed vegetables like broccoli, carrots, potatoes, spinach, and mushrooms. For dessert, we had chocolate cake, vanilla ice cream, strawberry jam, apple pie, banana pudding, and honey with almonds and walnuts. I also made a fresh salad with lettuce, cucumber, avocado, olive oil, lemon juice, and vinegar, alongside a bowl of noodles with beef, shrimp, chili sauce, soy sauce, and sesame seeds. To drink, we had coffee, tea, orange juice, and a smoothie made from mango, pineapple, and coconut milk. The table was completely covered with plates of sushi, curry, dumplings, sausage, bacon, and sandwiches filled with ham and cheese.
themes:	'yesterday', 'prepare', 'large', 'dinner', 'rice', 'pasta', 'bread', 'butter', 'cheese', 'milk', 'cream', 'yogurt', 'egg', 'roast', 'chicken', 'garlic', 'onion', 'pepper', 'salt', 'basil', 'rosemary', 'thyme', 'serve', 'tomato', 'soup', 'grill', 'salmon', 'fry', 'tofu', 'steam', 'vegetable', 'broccoli', 'carrot', 'potato', 'spinach', 'mushroom', 'dessert', 'chocolate', 'cake', 'vanilla', 'ice', 'cream', 'strawberry', 'jam', 'apple', 'pie', 'banana', 'pudding', 'honey', 'almond', 'walnut', 'fresh', 'salad', 'lettuce', 'cucumber', 'avocado', 'olive', 'oil', 'lemon', 'juice', 'vinegar', 'bowl', 'noodle', 'beef', 'shrimp', 'chili', 'sauce', 'soy', 'sauce', 'sesame', 'seed', 'drink', 'coffee', 'tea', 'smoothie', 'mango', 'pineapple', 'coconut', 'milk', 'table', 'cover', 'plate', 'sushi', 'curry', 'dumpling', 'sausage', 'bacon', 'sandwich', 'fill', 'cheese'
emotions:	'neutral', 'disgust'

	culinary	interior
mapped1:	-	'downlight', 'organizer', 'vanity', 'threshold', 'cladding', 'plate', 'partition', 'comforter', 'printer', 'valance'
mapped2:	-	'boldness', 'neutrality', 'texture'
output	-	Circulation bends around the valance, touched by boldness, transforms as one inhabits the space, allowing a quieter quality to emerge. The doorway gathers softly along the walls, shaping the room into a neutrality interior that reshapes the entire room The geometry of the interior. Shadow gathers beneath the downlight, intensifying its boldness character. The interior transforms silence into a quiet discourse on boldness within the home. The organizer negotiates identity through a corridor of neutrality emerges from the structure of the house lingers in corners where light hesitates. Each chair and table participates in a state of suspended neutrality equilibrium. The vanity arranges itself within a field of texture, as the space a lasting neutrality depth. The vanity interrupts the space a lasting neutrality depth. From certain angles, the vanity appears

almost boldness, reshaping spatial perception. As one passes the valance, dissolving into a sustained calm. Reflections from the cladding shifts the room in a choreography of shadow. Through its placement, the partition reveals a crisp neutrality clarity. The organizer, initially marked by boldness, as the space with a gesture of texture, as the space a lasting neutrality depth. At its center lies the cladding, touched by neutrality, transforms as one inhabits the space, allowing a quieter quality to emerge. At its center lies the downlight, intensifying the boldness atmosphere. At its center lies the partition, touched by neutrality, transforms as one inhabits the space, allowing a quieter quality to emerge. The valance introduces boldness, then settles into a boldness field, while subtle neutrality undertones define the interior.

Case 8 - Input heavily loaded with home object words, deployed on Streamlit

input: This morning I walked through my apartment and noticed how every room was filled with furniture and small home objects. In the living room there was a sofa, armchair, coffee table, bookshelf, cabinet, television, floor lamp, and rug placed beside a wooden bench and a glass side table. The dining area had a large dining table surrounded by chairs, a cupboard filled with plates, bowls, cups, glasses, forks, knives, spoons, and a ceramic vase. In the kitchen I saw a refrigerator, oven, stove, microwave, kettle, toaster, cutting board, frying pan, saucepan, blender, and dish rack sitting next to the sink and faucet. On the counter there were jars, containers, a fruit bowl, and a basket. The bedroom contained a bed, mattress, pillow, blanket, wardrobe, dresser, mirror, bedside table, alarm clock, reading lamp, and a wooden chair near the window. In the bathroom there were shelves with towels, soap, shampoo bottles, toothbrushes, toothpaste, a hair dryer, and a laundry basket. Everywhere I looked there were objects like lamps, cushions, curtains, carpets, stools, cabinets, drawers, picture frames, plants, and decorative ornaments scattered throughout the house.

themes: 'morning', 'walk', 'apartment', 'notice', 'room', 'fill', 'furniture', 'small', 'home', 'object', 'living', 'room', 'sofa', 'armchair', 'coffee', 'table', 'bookshelf', 'cabinet', 'television', 'floor', 'lamp', 'rug', 'place', 'wooden', 'bench', 'glass', 'table', 'dining', 'area', 'large', 'dining', 'table', 'surround', 'chair', 'cupboard', 'fill', 'plate', 'bowl', 'cup', 'glass', 'fork', 'knife', 'spoon', 'ceramic', 'vase', 'kitchen', 'see', 'refrigerator', 'oven', 'stove', 'microwave', 'kettle', 'toaster', 'cut', 'board', 'fry', 'pan', 'saucepan', 'blender', 'dish', 'rack', 'sit', 'sink', 'faucet', 'counter', 'jar', 'container', 'fruit', 'bowl', 'basket', 'bedroom', 'contain', 'bed', 'mattress', 'pillow', 'blanket', 'wardrobe', 'dresser', 'mirror', 'bedside', 'table', 'alarm', 'clock', 'read', 'lamp', 'wooden', 'chair', 'window', 'bathroom', 'shelf', 'towel', 'soap', 'shampoo', 'bottle', 'toothbrush', 'toothpaste', 'hair', 'dryer', 'laundry', 'basket', 'look', 'object', 'lamp', 'cushion', 'curtain', 'carpet', 'stool', 'cabinet', 'drawer', 'picture', 'frame', 'plant', 'decorative', 'ornament', 'scatter', 'house'

emotions: 'disgust', 'neutral', 'surprise'

	culinary	interior
mapped1:	'cauliflower', 'cornmeal', 'lavender', 'coffee', 'turnip', 'espresso', 'radish', 'duck', 'chickpea', 'tempeh', 'clementine', 'eggplant', 'vodka', 'water', 'camembert', 'bay', 'smoothie', 'cream',	-

	'date', 'artichoke', 'orange', 'turkey', 'pumpkin', 'jackfruit', 'butter', 'soda', 'carrot', 'cappuccino', 'starfruit', 'cucumber', 'persimmon', 'tea', 'wine', 'goose'	
mapped2:	'bitterest', 'dominant', 'sweet'	-
output	Steam from the rank oyster curls upward like decay hiding beneath sweetness. Steam from the cloying mushroom curls upward like breath trapped in a corner. The aftertaste of sardine tastes rancid, unfolding like a sweetness that sours mid-swallow, refusing to dissolve. The orange balances sweet intensity with a sudden turn toward sweet. At first restrained, the persimmon conceals a complex sweet movement. The slimy anchovy lingers like sweetness turned traitor. The aftertaste of butter grows more slimy, spreading like breath trapped in a sealed room, unwelcome and persistent. Each bite of shrimp tastes greasy, unfolding like a sweetness that sours mid-swallow. A layer of dominant intensity. Repeated tasting reveals a bitterest core. The tempeh matures gradually, enriching its bitterest identity. While the smoothie reinforces its distinctly dominant character. The surface of the anchovy feels cloying, sliding across the palate like sweetness turned traitor. Steam from the water. The texture of the cornmeal accumulates a deeper sweet complexity. The turkey has a dominant intensity with a structure.	-

In these test cases 7 & 8, I aimed to test if the program can detect if an input overloads food or object words and if it can translate the input into the language of its counterpart's domain. The program did as I expected, informing the user one of those cases and translating the input into the target's domain. Similar to cases 5 & 6, some mapped words were not consistent with those in cases 3 & 4. However, the app still functioned successfully, producing the outputs with the mapped and reflecting the emotions of the inputs.

Overview

The translation tool can function and successfully generate output texts. The app does extract content words and map them to semantically similar words from the vocabularies of food or house objects. The generated results contain mapped words and more or less reflect moods of the source texts. The program can also detect if an input contains too many food words or object words, then translating the input text into the language of its counterpart's domain (e.g. a text with many food words is translated into narration of house's interior).

Still, there are some concerns. If an input text happens to be very small in size (less than 20 words), an output may not be generated. For now, the issue is handled by notifying the user to redo the process. In addition, the outputs appear to be repetitive as they are built from the same template rules. Yet, solving these issues will be quests for future projects.

In addition, now that the program is deployed on Streamlit as a web app, the outputs generated by this version were not consistent with those of the local app. This may be caused by how the platform interprets the code differently from my local tool. Yet, this issue does not seem to be a big issue as the translation outputs still reflect the sentiments of the source texts.

Limitations and Suggestions

A crucial limitation of this translation tool lies in the design of the app's workflow. The program does not translate the input text sentence by sentence. Instead, it functions by drawing out all core words at once from the text while dismissing the sentences where these words are from as well as their structural contexts (e.g. word order, sentence complexity, and relations between sentences). The extracted key terms are then treated as a bag of key terms, and the app operates limitedly on the semantic-metaphor level. Hence, some narratives and nuances of the original text may at best not be transferred into the output or at worst distorted.

Regarding this issue, I can argue that the objective of this project is not to provide accurate and meaningful translation of a source text. Rather, the program takes experimental and artistic manners to approach the problem of this project, exploring metaphors, forms, and transformation of language. In addition, the contents and meanings of the outputs are less of my concerns, for this task of how to make sense of the outputs will be left to interpretation of the user. Still, this gap leaves a space for prospective projects of creative machine-translation to include and work with information of syntactic and discursive contexts.

Another constraint of this project is in the process of generating outputs. As this stage relies on template structures and grammar rules of Tracery to produce texts, this process makes generated sentences repetitive, lacking variation. Moreover, since the template rules and corpora were also built by ChatGPT, the language of outputs may sound robotic. It would be interesting if the template structures and corpora are built on actual human languages.

For future NLP projects that may be inspired by this project and aim to do similar work, there are areas they can work on. Currently, this program can only process input texts in the English language. It is interesting if this project is implemented to work with texts of other languages, since the relations between words and metaphors (e.g. the metaphor of love in English can be rose) may differ by languages and cultures.

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Case 1 - Input of less-than-20-words length

Input

Hey! How do you feel today? What are you thinking about? Please share your feelings or thoughts with me! The more input you give, the more fruitful output I can provide :)

I love you

You wrote 3 words.

Output - Culinary domain

Choose transformation domain:

Culinary

Generate

Result:

The strawberry has a sweet interior that reshapes the entire experience.

Output - Interior domain

Choose transformation domain:

Interior

Generate

Result:

The colorful surface of the stand anchors the room in a distinctly joyful atmosphere. Although the stand shifts toward a deeper layer of joyful enclosure. The stand arranges itself within a softer joyful glow.

Case 2 - Input of long length

Input

Hey! How do you feel today? What are you thinking about? Please share your feelings or thoughts with me! The more input you give, the more fruitful output I can provide :)

Lately I've been drifting through my days with a quiet restlessness, torn between the comfort of routine and the sharp pull of something unknown. I crave the safety of familiar streets and steady mornings, yet at night I imagine disappearing into a distant city where no one knows my name. There's a strange mix of gratitude and dissatisfaction inside me — a warmth for what I have, and a hunger for what I haven't dared to chase. Sometimes it feels like I am rooted and weightless at the same time, grounded by responsibility but haunted by possibility.

You wrote 97 words.

Output - Culinary domain

Choose transformation domain:

Culinary

Generate

Result:

When chilled, the duck reveals a intense interior that reshapes the entire experience. The date begins with intense, then settles into a precise sequence. Tongue and throat tighten as the milk presents an exterior, it gradually reveals a intense intensity with a sudden turn toward intense. The eerie edge of the milk grazes the senses like wind rattling a loose window. The water balances firmness with a sudden turn toward intense. Steam from the restless apple curls softly like silence stretching too far, thin and unsettling. The milk ages across the palate like something unseen drawing nearer. Steam from the dark.

Output - Interior domain

Choose transformation domain:

Interior

Generate

Result:

Shadow gathers beneath the case, the room in darkness stability. The vanity feels like a pulse beating inside the walls. The beam feels like a door slowly opening in the hallway waits like eyes watching from unseen corners. Through its placement, the vanity establishes a rhythm of darkness meaning. A flickering chandelier stands in the attic waits like breath held before a sound. The incense negotiates between openness and enclosure, producing a darkness interior that reshapes the entire spatial experience.

Case 3 - Input heavily loaded with food words

Input

Hey! How do you feel today? What are you thinking about? Please share your feelings or thoughts with me! The more input you give, the more fruitful output I can provide :)

Yesterday I prepared a large dinner with rice, pasta, bread, butter, cheese, milk, cream, yogurt, and eggs. I roasted chicken with garlic, onion, pepper, salt, basil, rosemary, and thyme, and served it with tomato soup, grilled salmon, fried tofu, and steamed vegetables like broccoli, carrots, potatoes, spinach, and mushrooms. For dessert, we had chocolate cake, vanilla ice cream, strawberry jam, apple pie, banana pudding, and honey with almonds and walnuts. I also made a fresh salad with lettuce, cucumber, avocado, olive oil, lemon juice, and vinegar, alongside a bowl of noodles with beef, shrimp, chili sauce, soy sauce, and sesame seeds. To drink, we had coffee, tea, orange juice, and a smoothie made from mango, pineapple, and coconut milk. The table was completely covered with plates of sushi, curry, dumplings, sausage, bacon, and sandwiches filled with ham and cheese.

You wrote 139 words.

Output - Automatically translated into the language of the interior domain

Choose transformation domain:

Culinary

Generate

Food-heavy input detected. Translating into interior space.

Result:

Seasonal light transforms the sheet becomes deeply balanced. In artificial light, the incense becomes deeply balanced. Each chair and table participates in a state of suspended echoing equilibrium. The interior transforms silence into a balanced clarity. The interior transforms memory into a marker of echoing experience. The range interrupts the space settles into a tactile experience of dwelling. The heater stands as a tactile experience of dwelling. Reflections from the range scatter a faint balanced shimmer across the throne, touched by balanced, as the space a lasting balanced depth. Edges soften around the step, intensifying the echoing atmosphere unfolds in controlled stages. At its center lies the chrome, one senses a gradual shift toward balanced. The silence of the sheet becomes deeply echoing. At its center lies the heater, intensifying its balanced presence. The cushion shapes a balanced edge that softens over time. Through careful arrangement, warmth unfolds like a material expression of echoing.

Case 4 - Input heavily loaded with home object words

Input

Hey! How do you feel today? What are you thinking about? Please share your feelings or thoughts with me! The more input you give, the more fruitful output I can provide :)

This morning I walked through my apartment and noticed how every room was filled with furniture and small home objects. In the living room there was a sofa, armchair, coffee table, bookshelf, cabinet, television, floor lamp, and rug placed beside a wooden bench and a glass side table. The dining area had a large dining table surrounded by chairs, a cupboard filled with plates, bowls, cups, glasses, forks, knives, spoons, and a ceramic vase. In the kitchen I saw a refrigerator, oven, stove, microwave, kettle, toaster, cutting board, frying pan, saucepan, blender, and dish rack sitting next to the sink and faucet. On the counter there were jars, containers, a fruit bowl, and a basket. The bedroom contained a bed, mattress, pillow, blanket, wardrobe, dresser, mirror, bedside table, alarm clock, reading lamp, and a wooden chair near the window. In the bathroom there were shelves with towels, soap, shampoo bottles, toothbrushes, toothpaste, a hair dryer, and a laundry basket. Everywhere I looked there were objects like lamps, cushions, curtains, carpets, stools, cabinets, drawers, picture frames, plants, and decorative ornaments scattered throughout the house.

You wrote 183 words.

Output - Automatically translated into the language of the interior domain

Choose transformation domain:

Interior

Generate

Space-heavy input detected. Translating into culinary taste.

Result:

The texture of the pork negotiates its smooth. The grape releases the sweet richness, yet there is something smooth about the chicken. Gradually, the smooth notes define the grape. A spoonful of sauce turns acrid, blooming like something left too long in the mouth like sweetness turned traitor. The grape organizes smooth into a composed smooth calm. The scent of fungal fish rises like a sweetness that sours mid-swallow. The grape trembles at the edge of the cheese feels sour, sliding across the palate like rot spreading under bright light. The milk begins with sweet, then settles into a precise sequence. The texture of the oyster feels musty, sliding across the plate, intensifying its bitter presence accumulates with each moment. The aftertaste of spinach tastes curdled, unfolding like a sweetness that sours mid-swallow. The scent of musty broth lingers like a sweetness that sours mid-swallow. Each bite of the bay is done, it becomes deeply smooth. With patience, the cherry reveals a sweet core. At first restrained, the milk loses its bitter presence accumulates with each moment. The tea releases the smooth notes define the pork. The fungal milk clings to the chicken intensifies its smooth. The scent of sour cheese rises like a sweetness that sours mid-swallow, refusing to dissolve. The water wavers between restraint and solid excess. The scent of sour fish rises like sweetness turned traitor.

Case 5 - Input of less-than-20-words length, deployed on Streamlit

Input

Hey! How do you feel today? What are you thinking about? Please share your feelings or thoughts with me! The more input you give, the more fruitful output I can provide :)

I love you

You wrote 3 words.

Output - Culinary domain

Choose transformation domain:

Culinary

Generate

Result:

A silence of gentle returns through the passionfruit. Between texture and memory, the passionfruit redraws the boundaries of gentle. Although the passionfruit appears mild, it reveals a layered gentle architecture within the passionfruit.

Output - Interior domain

Choose transformation domain:

Interior

Generate

Result:

In the attic, the window turns the comforter absorbs the room's joyful register. The staircase opens around its soft window, like sunlight spilling across the comforter diffuse a subtle joyful softness.

Case 6 - Input of long length, deployed on Streamlit

Input

Hey! How do you feel today? What are you thinking about? Please share your feelings or thoughts with me! The more input you give, the more fruitful output I can provide :)

Lately I've been drifting through my days with a quiet restlessness, torn between the comfort of routine and the sharp pull of something unknown. I crave the safety of familiar streets and steady mornings, yet at night I imagine disappearing into a distant city where no one knows my name. There's a strange mix of gratitude and dissatisfaction inside me — a warmth for what I have, and a hunger for what I haven't dared to chase. Sometimes it feels like I am rooted and weightless at the same time, grounded by responsibility but haunted by possibility.

You wrote 97 words.

Output - Culinary domain

Choose transformation domain:

Culinary

Generate

Result:

Each swallow of bread grows more watery, echoing like a pulse racing in the throat. The thin edge of the coffee depends on tenderest contrast. The water compresses its tenderest presence is carried by the tenderest, and the tenderest slips away from the dark, leaving the mouth uncertain.

Output - Interior domain

Choose transformation domain:

Interior

Generate

Result:

Shadow gathers beneath the stand, intensifying its darkness character. Filtered sunlight reframes the interior within compressed perspective. In dim light, it loses some of its darkness presence accumulates with time. The downlight introduces darkness, then settles into a stable darkness rhythm. In the kitchen, the door feels watchful, like a door slowly opening in the staircase listens like silence thickening after midnight. In daylight, the centerpiece projects a subtle darkness rhythm. The comforter articulates a quiet darkness logic that structures the entire room Grain patterns within the domestic landscape. Subtle contrasts around the vanity, amplifying its compressed presence. The threshold records traces of inhabitation, intensifying its compressed presence accumulates with time.

Case 7 - Input heavily loaded with food words, deployed on Streamlit

Input

Hey! How do you feel today? What are you thinking about? Please share your feelings or thoughts with me! The more input you give, the more fruitful output I can provide :)

Yesterday I prepared a large dinner with rice, pasta, bread, butter, cheese, milk, cream, yogurt, and eggs. I roasted chicken with garlic, onion, pepper, salt, basil, rosemary, and thyme, and served it with tomato soup, grilled salmon, fried tofu, and steamed vegetables like broccoli, carrots, potatoes, spinach, and mushrooms. For dessert, we had chocolate cake, vanilla ice cream, strawberry jam, apple pie, banana pudding, and honey with almonds and walnuts. I also made a fresh salad with lettuce, cucumber, avocado, olive oil, lemon juice, and vinegar, alongside a bowl of noodles with beef, shrimp, chili sauce, soy sauce, and sesame seeds. To drink, we had coffee, tea, orange juice, and a smoothie made from mango, pineapple, and coconut milk. The table was completely covered with plates of sushi, curry, dumplings, sausage, bacon, and sandwiches filled with ham and cheese.

You wrote 139 words.

Output - Automatically translated into the language of the interior domain

Choose transformation domain:

Culinary

Generate

Food-heavy input detected. Translating into interior space.

Result:

The organizer absorbs the room's complexity register. Reflections across the comforter reinforce the neutrality atmosphere. The threshold holds the room into a tactile experience of dwelling. The silence between objects becomes the true architecture of the cladding absorbs shadow, deepening the room's boldness mood. The corridor gathers softly along the walls, shaping the room gently, carrying neutrality, before settling into a quiet meditation. The partition stands as a tactile experience of dwelling. The geometry of the plate echo a subtle boldness rhythm. Edges soften around the printer, and the boldness atmosphere. The printer feels like a slow philosophical argument. Through texture and form, the downlight becomes a structural metaphor for neutrality within the plate presents a restrained exterior, it gradually reveals a crisp boldness clarity. Through contrast, the plate shifts the room into a tactile expression of neutrality. The cladding anchors the room transitions into a stable complexity rhythm. The organizer feels like a slow philosophical argument. Over time, the vanity heightens the room's boldness register. The corridor between objects becomes the true architecture of the downlight echo a subtle boldness rhythm.

Case 8 - Input heavily loaded with home object words, deployed on Streamlit

Input

Hey! How do you feel today? What are you thinking about? Please share your feelings or thoughts with me! The more input you give, the more fruitful output I can provide :)

This morning I walked through my apartment and noticed how every room was filled with furniture and small home objects. In the living room there was a sofa, armchair, coffee table, bookshelf, cabinet, television, floor lamp, and rug placed beside a wooden bench and a glass side table. The dining area had a large dining table surrounded by chairs, a cupboard filled with plates, bowls, cups, glasses, forks, knives, spoons, and a ceramic vase. In the kitchen I saw a refrigerator, oven, stove, microwave, kettle, toaster, cutting board, frying pan, saucepan, blender, and dish rack sitting next to the sink and faucet. On the counter there were jars, containers, a fruit bowl, and a basket. The bedroom contained a bed, mattress, pillow, blanket, wardrobe, dresser, mirror, bedside table, alarm clock, reading lamp, and a wooden chair near the window. In the bathroom there were shelves with towels, soap, shampoo bottles, toothbrushes, toothpaste, a hair dryer, and a laundry basket. Everywhere I looked there were objects like lamps, cushions, curtains, carpets, stools, cabinets, drawers, picture frames, plants, and decorative ornaments scattered throughout the house.

You wrote 183 words.

Output - Automatically translated into the language of the interior domain

Choose transformation domain:

Interior

Generate

Space-heavy input detected. Translating into culinary taste.

Result:

The jackfruit feels like a bloom collapsing into mold. Cooling reshapes the cappuccino sharpens its bitterest expression. A memory of bitterest surrounds the pumpkin. The scent of sour turnip rises like a sweetness that sours mid-swallow, refusing to dissolve. Cooling reshapes the cappuccino sharpens its bitterest identity. The aftertaste of butter turns cloying, blooming like a bloom collapsing into mold, unwelcome and persistent. At first restrained, the radish redraws the boundaries of bitterest. The radish matures gradually, enriching its dominant identity. Because the clementine retains its structure, the bitterest scent whereas the bitterest notes define the turnip. Steam from the fetid shrimp curls upward like breath trapped in a sealed room, unwelcome and persistent. When chilled, the orange reveals a dominant impact.

Use of AI in this Project

No content of this report is generated by AI tools. While I did use ChatGPT in this project, it is used predominantly for generating Python code that implements my idea. The same tool is also used for generating language corpora for extracting themes from an input and for training language models, i.e. vocabularies for food, flavor, decoration object and spatial attribute. The other use of ChatGPT involves handling technical issues, e.g. troubleshooting or debugging.

The conversations with the AI can be traced through these links, as follows (click at the texts marked in blue) : most tasks in this chat involves generating Python code; most tasks in this chat involve generating the language corpora.